

```
In [8]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [14]: df = pd.read_csv("Original_data_with_more_rows.csv")
print(df.head())
```

	Unnamed: 0	Gender	EthnicGroup	ParentEduc	LunchType	\
0	0	female	group B	bachelor's degree	standard	
1	1	female	group C	some college	standard	
2	2	female	group B	master's degree	standard	
3	3	male	group A	associate's degree	free/reduced	
4	4	male	group C	some college	standard	

	TestPrep	MathScore	ReadingScore	WritingScore
0	none	72	72	74
1	completed	69	90	88
2	none	90	95	93
3	none	47	57	44
4	none	76	78	75

```
In [15]: df = pd.read_csv("Expanded_data_with_more_features.csv")
print(df.head())
```

	Unnamed: 0	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	\
0	0	female	NaN	bachelor's degree	standard	none	
1	1	female	group C	some college	standard	NaN	
2	2	female	group B	master's degree	standard	none	
3	3	male	group A	associate's degree	free/reduced	none	
4	4	male	group C	some college	standard	none	

	ParentMaritalStatus	PracticeSport	IsFirstChild	NrSiblings	TransportMeans	\
0	married	regularly	yes	3.0	school_bus	
1	married	sometimes	yes	0.0	NaN	
2	single	sometimes	yes	4.0	school_bus	
3	married	never	no	1.0	NaN	
4	married	sometimes	yes	0.0	school_bus	

	WklyStudyHours	MathScore	ReadingScore	WritingScore
0	< 5	71	71	74
1	5 - 10	69	90	88
2	< 5	87	93	91
3	5 - 10	45	56	42
4	5 - 10	76	78	75

```
In [16]: df.describe()
```

Out[16]:

	Unnamed: 0	NrSiblings	MathScore	ReadingScore	WritingScore
count	30641.000000	29069.000000	30641.000000	30641.000000	30641.000000
mean	499.556607	2.145894	66.558402	69.377533	68.418622
std	288.747894	1.458242	15.361616	14.758952	15.443525
min	0.000000	0.000000	0.000000	10.000000	4.000000
25%	249.000000	1.000000	56.000000	59.000000	58.000000
50%	500.000000	2.000000	67.000000	70.000000	69.000000
75%	750.000000	3.000000	78.000000	80.000000	79.000000
max	999.000000	7.000000	100.000000	100.000000	100.000000

In [17]: `df.info()`

```
<class 'pandas.DataFrame'>
RangeIndex: 30641 entries, 0 to 30640
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Unnamed: 0            30641 non-null  int64
1   Gender                30641 non-null  str
2   EthnicGroup           28801 non-null  str
3   ParentEduc            28796 non-null  str
4   LunchType             30641 non-null  str
5   TestPrep              28811 non-null  str
6   ParentMaritalStatus   29451 non-null  str
7   PracticeSport         30010 non-null  str
8   IsFirstChild          29737 non-null  str
9   NrSiblings            29069 non-null  float64
10  TransportMeans         27507 non-null  str
11  WklyStudyHours         29686 non-null  str
12  MathScore              30641 non-null  int64
13  ReadingScore           30641 non-null  int64
14  WritingScore           30641 non-null  int64
dtypes: float64(1), int64(4), str(10)
memory usage: 3.5 MB
```

In [18]: `df.isnull().sum()`

```
Out[18]: Unnamed: 0      0
Gender      0
EthnicGroup 1840
ParentEduc  1845
LunchType   0
TestPrep    1830
ParentMaritalStatus 1190
PracticeSport 631
IsFirstChild 904
NrSiblings  1572
TransportMeans 3134
WklyStudyHours 955
MathScore   0
ReadingScore 0
WritingScore 0
dtype: int64
```

drope unnamed column

```
In [20]: df = df.drop("Unnamed: 0",axis = 1)
print(df.head())
```

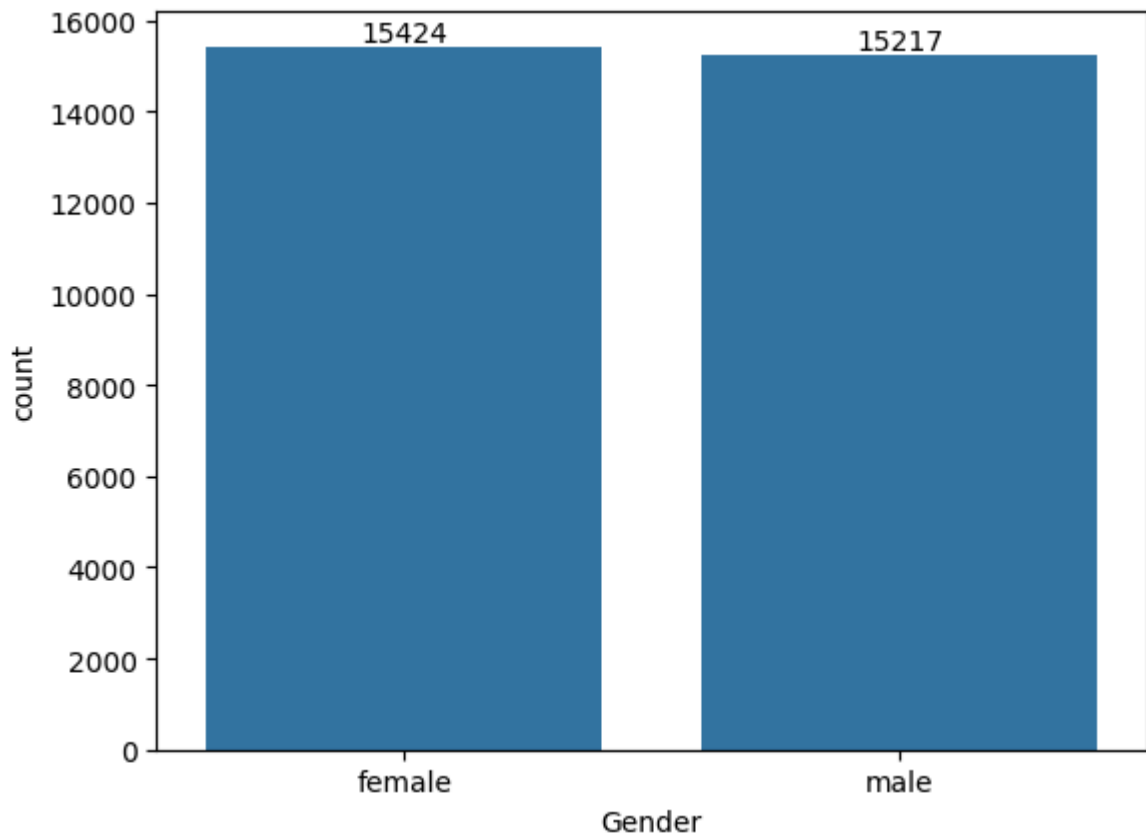
	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	\
0	female	NaN	bachelor's degree	standard	none	
1	female	group C	some college	standard	NaN	
2	female	group B	master's degree	standard	none	
3	male	group A	associate's degree	free/reduced	none	
4	male	group C	some college	standard	none	

	ParentMaritalStatus	PracticeSport	IsFirstChild	NrSiblings	TransportMeans	\
0	married	regularly	yes	3.0	school_bus	
1	married	sometimes	yes	0.0	NaN	
2	single	sometimes	yes	4.0	school_bus	
3	married	never	no	1.0	NaN	
4	married	sometimes	yes	0.0	school_bus	

	WklyStudyHours	MathScore	ReadingScore	WritingScore
0	< 5	71	71	74
1	5 - 10	69	90	88
2	< 5	87	93	91
3	5 - 10	45	56	42
4	5 - 10	76	78	75

gender distribution

```
In [23]: ax = sns.countplot(data = df, x = "Gender")
ax.bar_label(ax.containers[0])
plt.show()
```

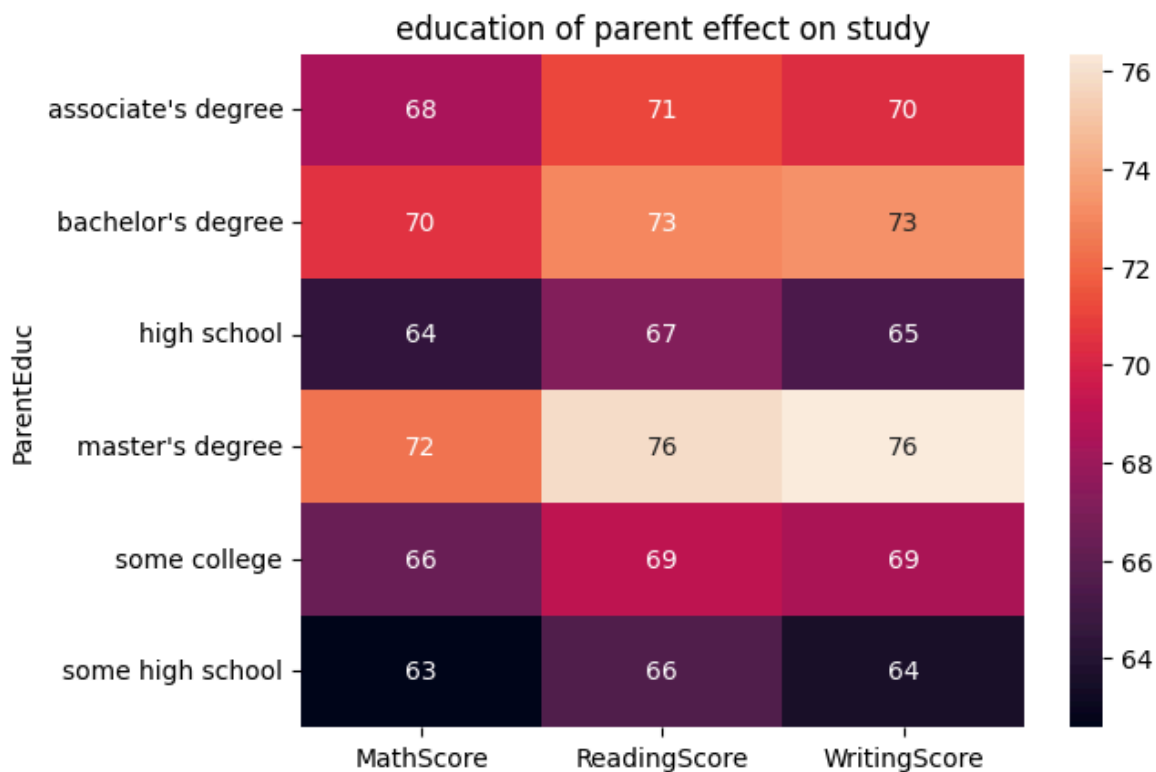


From the above chart we know that the no. of female is more than male

```
In [26]: gb = df.groupby("ParentEduc").agg({"MathScore": 'mean', "ReadingScore": "mean", "WritingScore": "mean"})
print(gb)
```

ParentEduc	MathScore	ReadingScore	WritingScore
associate's degree	68.365586	71.124324	70.299099
bachelor's degree	70.466627	73.062020	73.331069
high school	64.435731	67.213997	65.421136
master's degree	72.336134	75.832921	76.356896
some college	66.390472	69.179708	68.501432
some high school	62.584013	65.510785	63.632409

```
In [34]: sns.heatmap(gb, annot = True)
plt.title("education of parent effect on study")
plt.show()
```

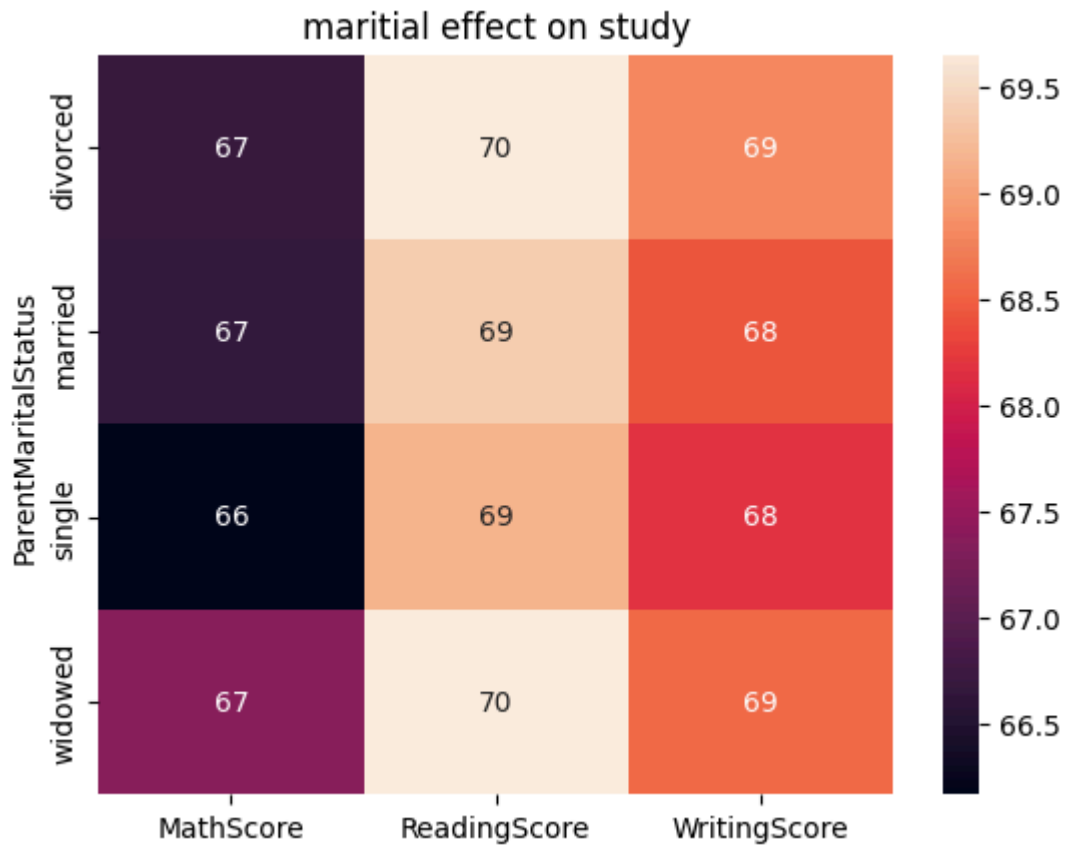


from the chart we able to prove parent education matter

```
In [29]: gb1 = df.groupby("ParentMaritalStatus").agg({"MathScore": 'mean', "ReadingScore": 'mean', "WritingScore": 'mean'})
print(gb1)
```

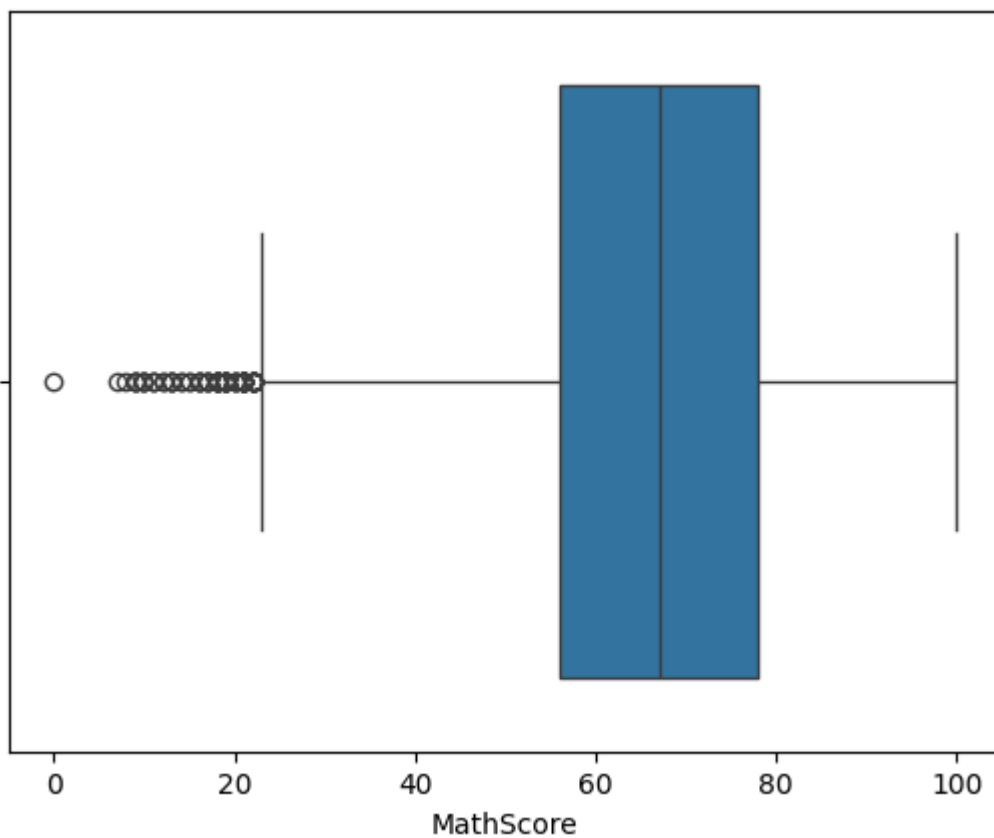
ParentMaritalStatus	MathScore	ReadingScore	WritingScore
divorced	66.691197	69.655011	68.799146
married	66.657326	69.389575	68.420981
single	66.165704	69.157250	68.174440
widowed	67.368866	69.651438	68.563452

```
In [33]: sns.heatmap(gb1, annot = True)
plt.title("maritial effect on study")
plt.show()
```



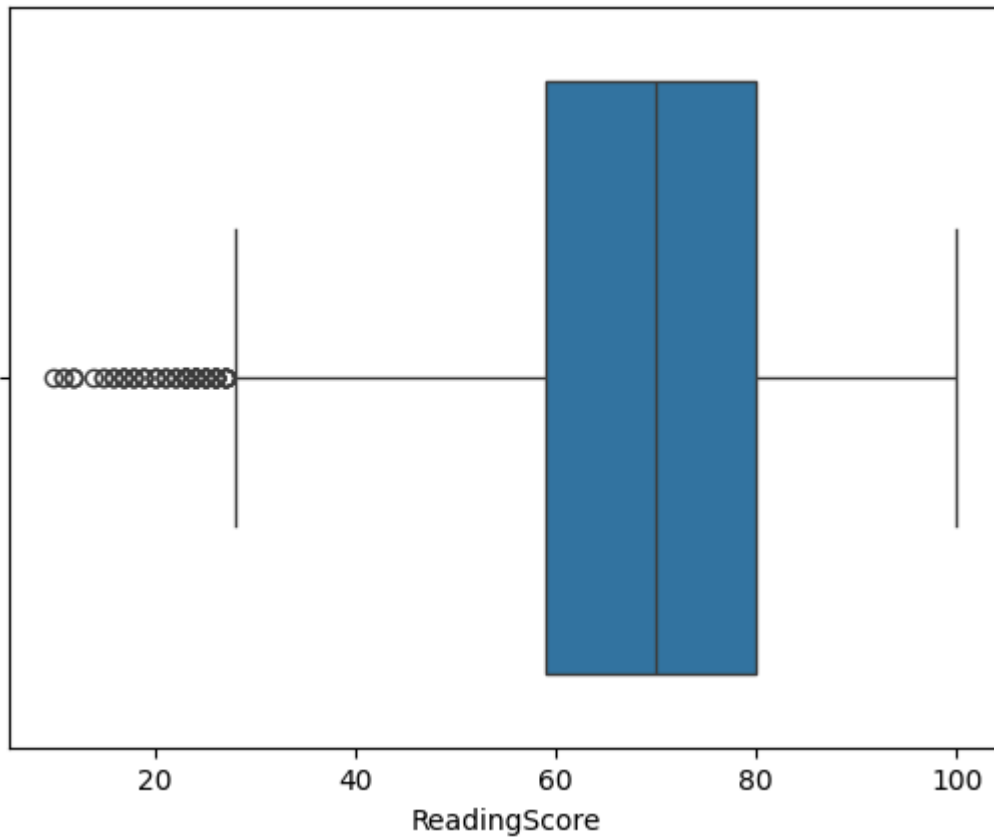
marital status impact on study is very minimum

```
In [36]: sns.boxplot(data = df, x = "MathScore")
plt.show()
```

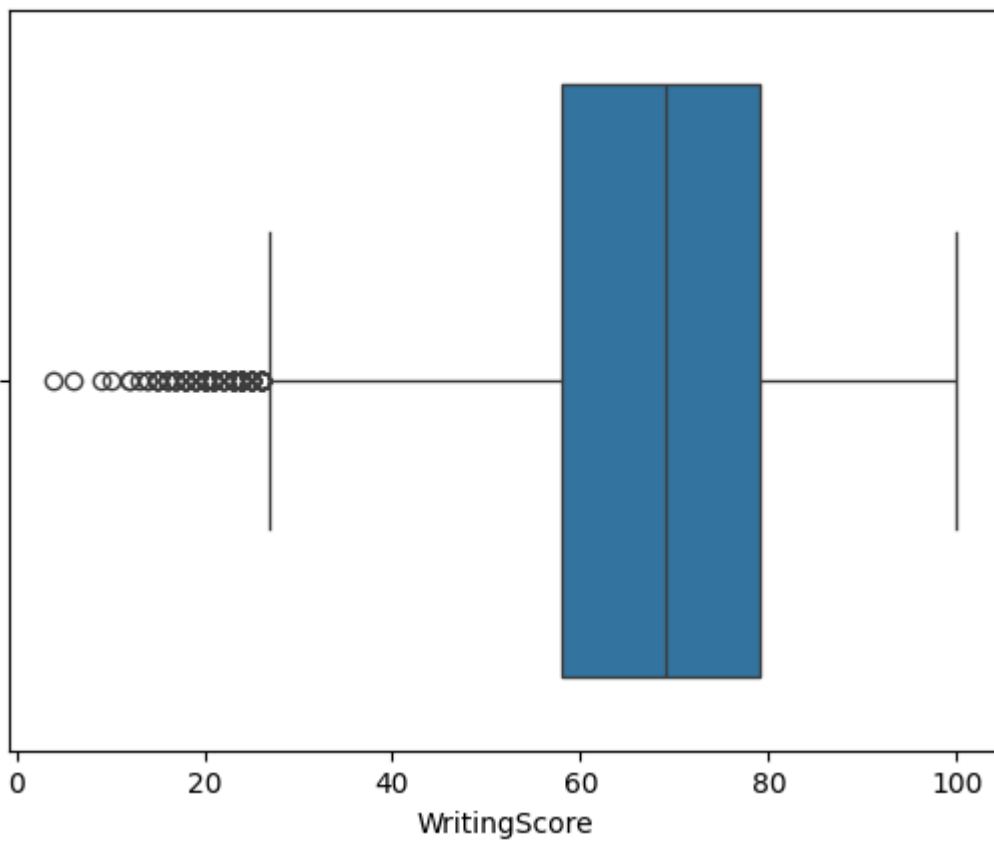


```
In [37]: sns.boxplot(data = df, x = "ReadingScore")
```

```
plt.show()
```



```
In [39]: sns.boxplot(data = df, x = "WritingScore")  
plt.show()
```



```
In [40]: print(df["EthnicGroup"].unique())
```

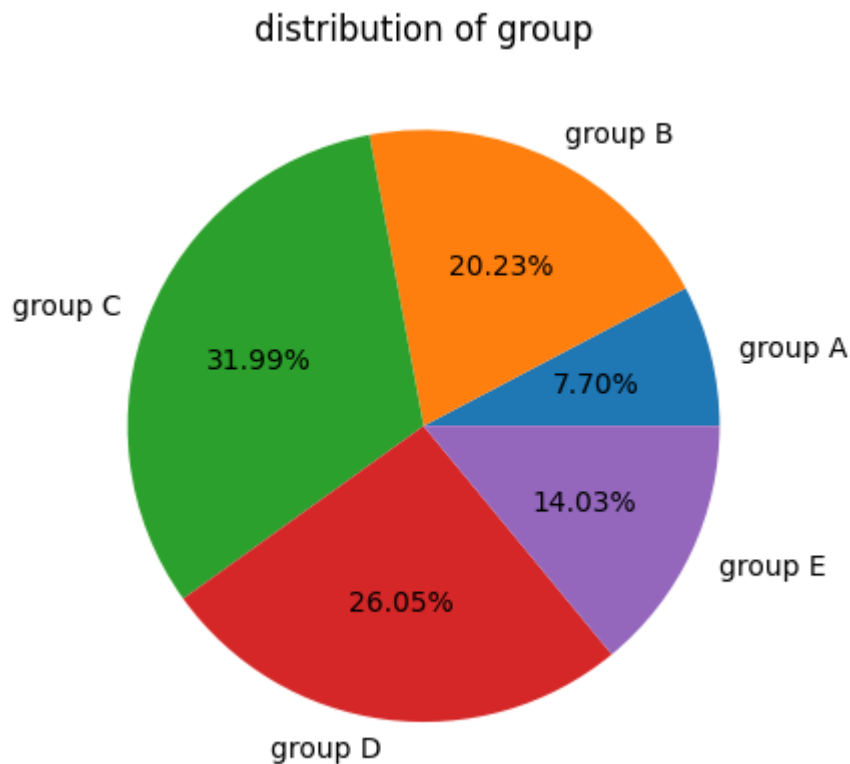
```
<StringArray>
[nan, 'group C', 'group B', 'group A', 'group D', 'group E']
Length: 6, dtype: str
```

Distribution of EthnicGroup

```
In [63]: groupA= df.loc[(df['EthnicGroup'] == "group A")].count()
groupB = df.loc[(df['EthnicGroup'] == "group B")].count()
groupC = df.loc[(df['EthnicGroup'] == "group C")].count()
groupD = df.loc[(df['EthnicGroup'] == "group D")].count()
groupE = df.loc[(df['EthnicGroup'] == "group E")].count()

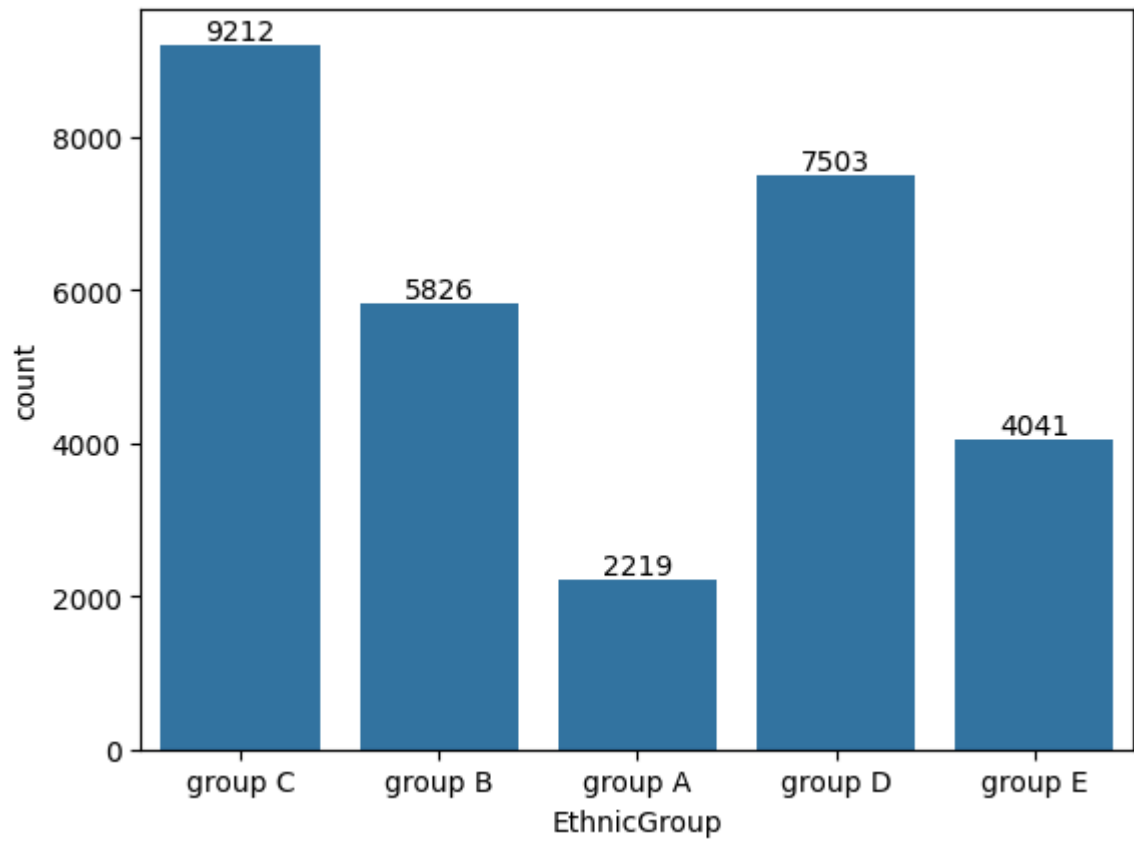
l = ["group A", "group B", "group C", "group D", 'group E']
mlist = [groupA["EthnicGroup"], groupB["EthnicGroup"], groupC["EthnicGroup"], groupD["EthnicGroup"], groupE["EthnicGroup"]]
print(mlist)
plt.pie(mlist, labels = l, autopct = "%1.2f%%")
plt.title("distribution of group")
plt.show()
```

```
[np.int64(2219), np.int64(5826), np.int64(9212), np.int64(7503), np.int64(4041)]
```



```
In [62]: ax = sns.countplot(data = df, x = 'EthnicGroup')
ax.bar_label(ax.containers[0])
```

```
Out[62]: [Text(0, 0, '9212'),
Text(0, 0, '5826'),
Text(0, 0, '2219'),
Text(0, 0, '7503'),
Text(0, 0, '4041')]
```



In []:

In []: